

Manikanta Reddy Duggempudi

Data Analyst

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Professional Summary

Experienced Data Analyst with 3+ years of expertise in predictive modeling, machine learning, and advanced analytics across healthcare, finance, and supply chain domains. Proficient in Python, SQL, Spark, Databricks, Snowflake, Azure Data Factory, Tableau, and Power BI for building predictive models, scalable ETL pipelines, and interactive dashboards. Skilled in time-series forecasting, clustering, anomaly detection, NLP, and survival analysis to extract insights from complex clinical, claims, and financial datasets. Proven track record of reducing hospital readmissions, optimizing drug inventory, cutting fraud losses, and improving operational efficiency. Adept at collaborating with cross-functional teams to translate data into strategies that drive cost savings, better patient outcomes, and stronger business performance.

Skills

Data Analysis & Modeling: EDA, Statistical Analysis, Time-Series Forecasting, Logistic Regression, Random Forest, Gradient Boosting, Clustering, Anomaly Detection, Survival Analysis

Data Engineering & Management: SQL, ETL Development, Data Wrangling, Snowflake, AWS (S3, DMS, EC2), Azure Data Factory, BigQuery, Palantir Foundry, Databricks, Spark (PySpark)

NLP & Document Processing: spaCy, TensorFlow, NER, Text Classification, OCR, Speech-to-Text, Intent Classification, Clinical Notes & Claims Analytics

Visualization & Reporting: Power BI, Tableau, Matplotlib, Seaborn, Dashboard Development, Automated Reporting

Programming & Tools: Python (Pandas, NumPy, Scikit-learn), SQL, Excel (VBA, Power Query), RPA (iD2E), MLOps

Forecasting & Predictive Analytics: Demand Forecasting, Inventory Optimization, Occupancy Prediction, Excel Forecast Sheets

Reporting & Automation: Power BI Report Automation, Excel VBA, Automated Data Pipelines

Domain Expertise: Healthcare, Supply Chain & Inventory, Finance & Lending, Retail & Marketing Analytics

Experience

Cardinal Health, USA

Feb 2025 – Present

Data Analyst

- Built predictive inventory models with Python, SQL, Spark, and Databricks on hospital orders, inventory, and shipment data, reducing stockouts and cutting excess inventory by double digits.
- Designed ETL pipelines in Azure Data Factory, Palantir Foundry, and BigQuery to integrate order logs, supplier schedules, and patient usage, enabling real-time forecasting.
- Applied time-series (ARIMA, LSTM) and gradient-boosted trees, delivering accurate demand forecasts and saving \$8.25M in 3 months through optimized purchasing.
- Created Tableau and Power BI dashboards for supply chain leaders, improving visibility into demand, savings, and fill-rate metrics.
- Developed readmission-risk models with logistic regression, Random Forest, and gradient boosting on EHR, claims, and lab data, reducing 30-day readmissions.
- Built NLP workflows in spaCy and TensorFlow to analyze physician notes and clinical documents, extracting phenotypes and care gaps for better treatment planning.
- Performed clustering and survival analysis on clinical and pharmacy data to improve adherence monitoring and guide personalized care interventions.
- Partnered with clinical and operations teams to turn analytics into strategies, optimizing supply chain, improving outcomes, and increasing trial enrollment.

Hexaware Technology, India

May 2020 – Nov 2022

Data Analyst

- Consolidated 35 TB of siloed loan and transaction data into a centralized data warehouse using AWS (S3, DMS, EC2), Snowflake, and SQL Server, enabling advanced analytics across the lending portfolio.
- Built fraud detection and churn prediction models in Python and Spark ML, applying anomaly detection, classification, and regression on customer transactions and behavior to reduce losses and improve retention.
- Developed an MLOps pipeline for continuous model training and deployment, ensuring robust monitoring and scalability of fraud scoring and credit risk prediction systems.
- Delivered Tableau dashboards integrated with Snowflake, providing real-time visibility into lending risk metrics, customer churn insights, and operational performance.
- Designed NLP and OCR pipelines in Python to auto-index and classify ~4 million faxed patient records, cutting medical record processing time by 50% and reducing manual workload.
- Deployed speech-to-text and intent classification workflows for inbound patient calls, improving nurse-triage response efficiency by 40% and reducing call resolution times by 25%.
- Implemented predictive analytics on claims data integrated with Epic EHR, identifying potential rejections and reducing insurance claim denials by 25%, accelerating reimbursement cycles.
- Automated repetitive revenue cycle and data validation tasks using RPA (iD2E) with Python and SQL integration, lowering operational costs by 20% and boosting patient satisfaction scores.

Education

Masters in Information Technology and Management, University of North Carolina, NC, USA

Dec 2024